Assignment 12: Conditionals

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Assignment

1. Choice Operators (10 pts)

1a

```
[1] TRUE
```

1b

```
[1] TRUE
```

1c

```
[1]  TRUE FALSE FALSE  TRUE  TRUE
```

1d

```
[1] TRUE
```

1e

```
[1] TRUE
```

2. Using `dplyr` Choice Functions (20 points)

2a

```
[1] "small"
```

2b

```
[1] "medium"
```

2c

```
# A tibble: 392 x 10
   date       latitude site   size size_category air_temp air_temp_sd water_temp
   <date>        <dbl> <chr> <dbl> <chr>            <dbl>       <dbl>      <dbl>
 1 2016-07-24       30 GTM    12.4 small             21.8        6.39       24.5
 2 2016-07-24       30 GTM    14.2 small             21.8        6.39       24.5
 3 2016-07-24       30 GTM    14.5 small             21.8        6.39       24.5
 4 2016-07-24       30 GTM    12.9 small             21.8        6.39       24.5
 5 2016-07-24       30 GTM    12.4 small             21.8        6.39       24.5
 6 2016-07-24       30 GTM    13.0 small             21.8        6.39       24.5
 7 2016-07-24       30 GTM    10.3 small             21.8        6.39       24.5
 8 2016-07-24       30 GTM    11.2 small             21.8        6.39       24.5
 9 2016-07-24       30 GTM    12.7 small             21.8        6.39       24.5
10 2016-07-24       30 GTM    14.6 small             21.8        6.39       24.5
# i 382 more rows
# i 2 more variables: water_temp_sd <dbl>, name <chr>
```

2d

```
# A tibble: 392 x 10
   date       latitude site   size size_category3 air_temp air_temp_sd
   <date>        <dbl> <chr> <dbl> <chr>             <dbl>       <dbl>
 1 2016-07-24       30 GTM    12.4 medium            21.8        6.39
 2 2016-07-24       30 GTM    14.2 medium            21.8        6.39
 3 2016-07-24       30 GTM    14.5 medium            21.8        6.39
 4 2016-07-24       30 GTM    12.9 medium            21.8        6.39
 5 2016-07-24       30 GTM    12.4 medium            21.8        6.39
 6 2016-07-24       30 GTM    13.0 medium            21.8        6.39
 7 2016-07-24       30 GTM    10.3 medium            21.8        6.39
```

```
 8 2016-07-24       30 GTM    11.2 medium             21.8        6.39
 9 2016-07-24       30 GTM    12.7 medium             21.8        6.39
10 2016-07-24       30 GTM    14.6 medium             21.8        6.39
# i 382 more rows
# i 3 more variables: water_temp <dbl>, water_temp_sd <dbl>, name <chr>
```

3. If/Else Statements (20 points)

3a

```
[1] 0
```

3b

```
[1] 10
```

```
[1] 5
```

```
[1] 0
```

```
[1] 0
```

```
[1] 0
```

3c

```
[1] 10
```

```
[1] 24.5
```

```
[1] NA
```

4. Size Estimates by Name (20 points)

4a

```
[1] 4779.848
```

4b

```
[1] 1385.286
```

4c

```
[1] 8070.685
```

4d

```
[1] NA
```

Challenge 1 (optional)

```
Warning in get_mass_from_length_by_name(13, "Ankylosauria"): No known
estimation for Ankylosauria
```

```
[1] NA
```

Challenge 2 (optional)

```
[1] 1283.047
```

5. Putting It All Together: Portal Rodents (30 points)

5a

```
# A tibble: 6 x 13
  record_id month   day  year plot_id species_id sex   hindfoot_length weight
      <dbl> <dbl> <dbl> <dbl>   <dbl> <chr>      <chr>           <dbl>  <dbl>
1         1     7    16  1977       2 NL         M                  32     NA
2         2     7    16  1977       3 NL         M                  33     NA
3         3     7    16  1977       2 DM         F                  37     NA
4         4     7    16  1977       7 DM         M                  36     NA
5         5     7    16  1977       3 DM         M                  35     NA
6         6     7    16  1977       1 PF         M                  14     NA
# i 4 more variables: genus <chr>, species <chr>, taxa <chr>,
#   scientific_name <chr>
```

5b

```
# A tibble: 2 x 2
  id_level       n
  <chr>      <int>
1 genus only    70
2 species    34177
```

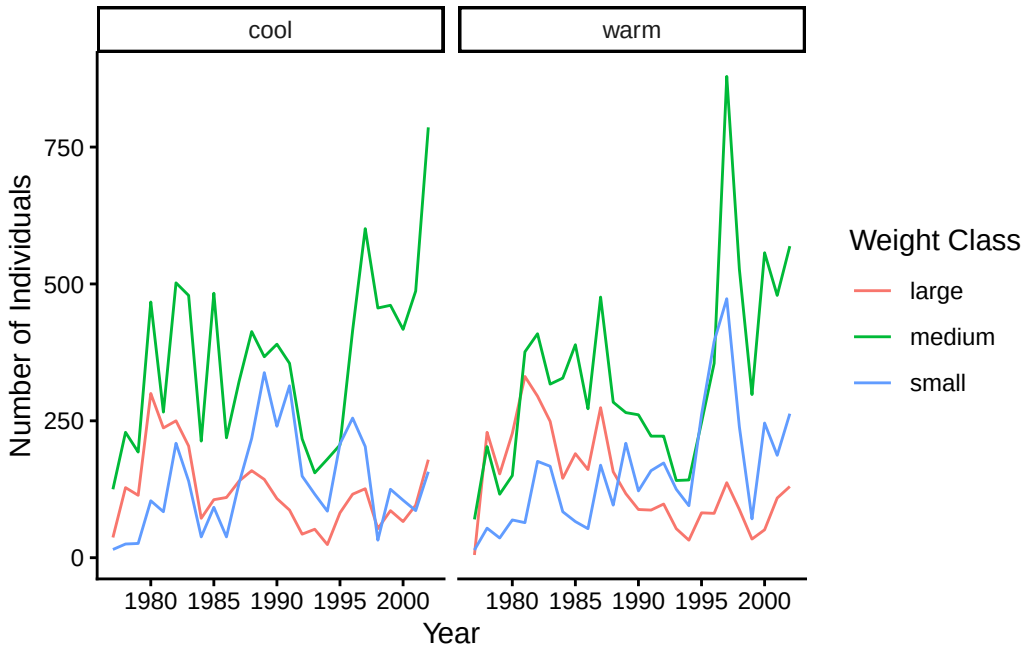
5c

5d

```
# A tibble: 4 x 2
  weight_class     n
  <chr>        <int>
1 large         6721
2 medium       17958
3 small         7604
4 unknown       1964
```

Without the "unknown" line, individuals with missing weights would be classified as "large." `weight < 20` and `weight < 50` both return `NA` when `weight` is `NA`, and `case_when` treats `NA` like `FALSE`, so those rows fall all the way through to the `.default` value.

5e



Answers will vary. Any accurate description of a pattern in the plot (e.g., which weight class is most common, how counts change over time, or differences between seasons) is acceptable.

6. Challenge: A "Safe" Tidyverse Function (OPTIONAL)

6a

```
# A tibble: 34,247 x 3
   species_id weight category
   <chr>       <dbl> <chr>
 1 NL             NA <NA>
 2 NL             NA <NA>
 3 DM             NA <NA>
 4 DM             NA <NA>
 5 DM             NA <NA>
 6 PF             NA <NA>
 7 PE             NA <NA>
 8 DM             NA <NA>
 9 DM             NA <NA>
10 PF             NA <NA>
# i 34,237 more rows
```

6b

```
# A tibble: 34,247 x 3
   species_id hindfoot_length category
   <chr>                <dbl> <chr>
 1 NL                      32 above
 2 NL                      33 above
 3 DM                      37 above
 4 DM                      36 above
 5 DM                      35 above
 6 PF                      14 below
 7 PE                      NA <NA>
 8 DM                      37 above
 9 DM                      34 above
10 PF                      20 below
# i 34,237 more rows
```

6c

```
Error in `categorize_by_threshold()`:
! Column must be numeric.
```

6d

```
Error in `categorize_by_threshold()`:
! Threshold must be between 4 and 280
```